# Xuanzhi Chen

+86 139-6877-5028 | xuanzhichen@stu.pku.edu.cn
https://xuanzchen.github.io | https://github.com/xuanzchen
https://scholar.google.com/citations?user=rc119DYAAAAJ | https://www.linkedin.com/in/xuanzhi-chen-9a0698408/

## Research Interests

Spatial multi-omics, machine learning for biology, and scientific agents.

## Education

**Peking University** Beijing, China
*B.Eng. Candidate in Bioinformatics, School of Life Sciences* *Sep. 2023 – Jun. 2027 (Expected)*

- **GPA: 3.77/4.0.** Pursuing a double degree in Economics at the National School of Development.
- **Selected coursework:** Data Structures and Algorithms (98/100); Statistical Analysis of Genomics Data (A).

## Preprint

Nianping Liu, **Xuanzhi Chen**, Miao Cui, Xiaoying Liao, Xiaoke Song, Sairam Pantham, Weize Xu, and Xiaojie Qiu. **Bio-Babel: autonomous cross-language reconstruction of agent-ready computational biology ecosystems.** *bioRxiv*, 2026. https://doi.org/10.64898/2026.08.30.743167.

## Research Experience

**College of Future Technology, Peking University** Beijing, China
*Undergraduate Researcher; Advisor: Prof. Hu Zeng* *Jul. 2024 – Present*

- **STATES-GBM:** Optimized stitching and segmentation for glioblastoma spatial multi-omics: seams misaligned by $> 3$ pixels fell from 14.2% to 0.3%, and multi-nucleus segments from 28.8% to 13–14% (common 3D nuclear reference). Analyzed malignant-cell identities, subtypes, and spatial translation efficiency.
- **iST Agent:** Developed a contract- and memory-guided agent for imaging-based spatial transcriptomics. Generated machine-readable contracts for seven pipelines and reproduced a laboratory pipeline's decoded output byte for byte.

**Department of Genetics, Stanford University** Stanford, CA
*Undergraduate Researcher; Advisor: Prof. Xiaojie Qiu* *Jun. 2026 – Present*

On-site summer research, Jun.–Aug. 2026; continued remotely.

- **Bio-Babel:** Built an agent to generate machine-readable usage contracts for reconstructed packages; expanded coding-agent support and tooling for agent-driven Python-to-C++ reconstruction. Conducted the pancreatic differentiation case study using Bio-Babel's analysis agents; reviewed their outputs and completed genotype-aware screening and biological interpretation of reduced maturation signatures under SWI/SNF loss. https://github.com/Bio-Babel/bio-babel-pancreas | https://github.com/Bio-Babel/UMI-tools-cpp
- **End-to-end spatial transcriptomics modeling:** Developing and training a model on public MERFISH data for joint molecular localization, cross-round correspondence, barcode decoding, cell segmentation, and molecule-to-cell assignment using shared fluorescence and morphology representations.

**Canchen Technology** Remote
*Research Intern; Mentor: Dr. Lei Li* *Oct. 2026 – Present*

- Research focus: agentic world models for virtual tissue.

## Presentation

**Bio-Babel.** Invited oral presentation, PKU Undergraduate Honors Program in Biology Annual Symposium, Oct. 2026.

## Honors and Awards

Academic Excellence Award, Peking University 2026
Peking University Scholarship 2025
Peking University Freshmen Scholarship 2023
Gold Medal, China National Biology Olympiad Final 2022

## Technical Skills

**Experimental:** In situ spatial omics (STATES), molecular cloning, CRISPR-Cas9, PCR/qPCR, Western blotting.
**Computational:** Python, R; single-cell and spatial omics, image analysis, scientific agents, workflow automation.